Extending the Framework: Triple-Quad Function on a 3-Folded Packet

Research & Development: BioChain AI

Subject: Projective Geometric Topological Data Analytics & Inverse O(n³) Reconstruction

To make this mathematically coherent, we define the **triple-quad function**, the **3-folded packet**, and the way projective geometric topological data analytics can be used to reverse engineer an unknown value from an $O(n^3)$ interaction structure.

1. From Four Folds to Three Folds

Previously, the four-fold packet gave

$$2^4 = 16$$

layers and planar footprint ratio

$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

Now suppose we stop after **three folds**.

If each fold halves the planar footprint and doubles the layer count, then after three folds we get

$$A_3 = \frac{A_0}{2^3} = \frac{A_0}{8},$$

and

$$L_3 = 2^3 = 8.$$

So a **3-folded packet** has:

- planar footprint ratio $1/8$,
- 8 stacked layers,
- dyadic layer symmetry,
- enough internal structure to encode higher-order interactions.

Thus, the 3-folded packet is the natural intermediate object between the original square and the fully stabilized 16 -layer packet.

2. Why a 3-Folded Packet Is Linked to Cubic Scaling

A square data manifold has

$$n^2$$

sites. A purely planar analysis sees only $O(n^2)$ structure.

To obtain

$$O(n^3),$$

we need one additional effective dimension. In the folded-packet model, that dimension is not necessarily physical depth alone. It can also be:

- layer depth,
- projection depth,
- resonance mode depth,
- cross-layer adjacency depth,
- filtration depth in topological data analysis.

So the central principle is:

$$\text{planar data} \times \text{ depth parameter} = O(n^2) \cdot O(n) = O(n^3).$$

The 3-folded packet supplies the **local layered structure**, while the projective/topological analysis supplies the **global depth parameter**.

3. Defining the Triple-Quad Function

We need a precise definition. Let the original square data field be

$$X : \{1, \dots, n\} \times \{1, \dots, n\} \to \mathbb{R}.$$

After three folds, each visible projected coordinate contains up to 8 stacked layer values. Let these layer values be

$$x_1, x_2, \ldots, x_8.$$

A **quad function** should measure a quadratic interaction among four related data sites. So define a local quad functional

$$Q(a,b,c,d) = \alpha_1 a^2 + \alpha_2 b^2 + \alpha_3 c^2 + \alpha_4 d^2 + \beta_{12} ab + \beta_{13} ac + \beta_{14} ad + \beta_{23} bc + \beta_{24} bd + \beta_{34} cd.$$

This is the most general quadratic energy-like function on four variables.

A **triple-quad function** then combines three such quad interactions:

$$\mathcal{TQ}(X) = Q_1(X_A) + Q_2(X_B) + Q_3(X_C),$$

where X_A, X_B, X_C are three four-site projections selected from the folded packet.

For example, if the eight folded layers are $x_1, x_2, \ldots, x_8$, one possible triple-quad functional is

$$\mathcal{TQ}(x_1, \ldots, x_8) = Q_1(x_1, x_2, x_3, x_4) + Q_2(x_3, x_4, x_5, x_6) + Q_3(x_5, x_6, x_7, x_8).$$

The overlaps are important. They allow information to propagate between layers. So the triple-quad function is a **three-channel quadratic measurement system** on the folded packet.

4. Why "Triple-Quad" Naturally Suggests $O(n^3)$

There are three structural levels:

1. **Planar index**

$$i = 1, \ldots, n.$$

2. **Planar index**

$$j = 1, \ldots, n.$$

3. **Depth, mode, or projection index**

$$k = 1, \ldots, n.$$

Together, these form a cubic index space:

$$(i, j, k) \in \{1, \ldots, n\}^3.$$

Thus the full interaction tensor has size

$$n \times n \times n,$$

so the total number of possible indexed interactions is

$$n^3.$$

The triple-quad function is a way to sample or reconstruct this cubic interaction field through local quadratic measurements. We can write the reconstructed cubic field as

$$V(i, j, k).$$

Then the total topological interaction cost is

$$C_{\text{top}}(n) = \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} \Gamma(i, j, k),$$

where $\Gamma(i, j, k)$ is the interaction weight at planar site (i, j) and depth level k. Therefore,

$$C_{\text{top}}(n) = O(n^3).$$

5. Projective Geometric Interpretation

Now we connect this to **projective geometry**.

In projective geometry, points are studied up to scale. A point in projective space is represented as

$$[x : y : z : w],$$

where

$$[x : y : z : w] = [\lambda x : \lambda y : \lambda z : \lambda w]$$

for every nonzero scalar λ.

A folded packet naturally creates multiple representatives of the same underlying data point because several layers project onto the same planar footprint. So the 3-folded packet can be modeled as a projective bundle:

$$\pi : \mathcal{P} \to S,$$

where:

- $\mathcal{P}$ is the folded packet,
- S is the original square base,
- π is the projection map from folded layers back onto the planar square.

For each base point (i, j), the fiber is

$$\pi^{-1}(i, j).$$

After three folds, this fiber contains up to 8 layer representatives:

$$\pi^{-1}(i, j) = \{x_1, \ldots, x_8\}.$$

The triple-quad function measures quadratic relationships inside this fiber and between neighboring fibers.

6. Topological Data Analytics Layer

Topological data analysis studies how structure persists across scales. We introduce a filtration parameter ϵ.

At each scale ϵ, we build a complex K_ϵ. For example, K_ϵ may be a Vietoris-Rips complex, alpha complex, or cubical complex built from the folded packet. As ϵ increases, the topology changes:

$$K_{\epsilon_1} \subseteq K_{\epsilon_2} \subseteq K_{\epsilon_3} \subseteq \cdots.$$

Persistent homology measures features such as connected components, loops, voids, and

cavities. These are measured by Betti numbers:

$$\beta_0, \beta_1, \beta_2, \ldots$$

For a folded toroidal panmagic chain, the most relevant quantities are usually:

- β_0 : number of connected components,
- β_1 : number of independent loops,
- β_2 : number of enclosed voids or cavities.

The topological signature may be written as

$$\mathcal{B}(\epsilon) = (\beta_0(\epsilon), \beta_1(\epsilon), \beta_2(\epsilon)) .$$

7. Reverse Engineering the Value

Now we formulate what “reverse engineer the value” means.

Suppose there is an unknown hidden value or field $V(i,j,k)$ inside the cubic interaction structure. We do not observe V directly. Instead, we observe:

1. projected planar data,
2. folded layer interactions,
3. triple-quad measurements,
4. persistent topological signatures.

Let the observed data be Y. The forward model is

$$Y = \mathcal{F}(V),$$

where $\mathcal{F}$ is the combined operation:

$$\mathcal{F} = \mathrm{TDA} \circ \mathrm{Projection} \circ \mathrm{TripleQuad} \circ \mathrm{Folding}.$$

So explicitly,

$$Y = \mathrm{TDA}\left(\Pi\left(\mathcal{TQ}\left(F_3(V)\right)\right)\right).$$

Here:

- F_3 is the three-fold packet transformation,
- $\mathcal{TQ}$ is the triple-quad measurement,
- Π is the projective projection,
- TDA extracts topological features.

Reverse engineering means solving the inverse problem:

$$V^* = \arg\min_V \|\mathcal{F}(V) - Y\|^2 + \lambda R(V).$$

Here:

- V^* is the reconstructed value,
- $R(V)$ is a regularization term,
- $\lambda > 0$ controls stability.

The regularization should encode golden subdivision and anti-resonance constraints.

8. Golden Regularization

Because the model is based on golden subdivision, the reconstructed value should penalize rational periodic alignment. Let $G_\phi(V)$ be a golden-ratio regularizer. One possible form is

$$G_\phi(V) = \sum_{p,q} w_{pq} \left|\langle V, \psi_p \rangle\right|^2 \chi_{\mathrm{res}}(p,q),$$

where:

- ψ_p are Fourier or wavelet modes,
- $\chi_{\mathrm{res}}(p,q)$ is large when the mode is near a rational resonance,
- w_{pq} weights the resonance strength.

Then the inverse problem becomes

$$V^* = \arg\min_V \left[\|\mathcal{F}(V) - Y\|^2 + \lambda_1 R(V) + \lambda_2 G_\phi(V)\right].$$

This dictates:

- match the observed projective-topological data,
- keep the reconstruction geometrically smooth,
- suppress resonant rational alignments.

9. The Role of the Triple-Quad Function in Reconstruction

The triple-quad function is important because it transforms the folded packet into a measurable energy object. Instead of trying to reconstruct raw layer values directly, we reconstruct them through quadratic invariants.

For a local folded fiber $(x_1, \dots, x_8)$, the triple-quad output is

$$\mathcal{TQ}(x_1, \dots, x_8) = Q_1(x_1, x_2, x_3, x_4) + Q_2(x_3, x_4, x_5, x_6) + Q_3(x_5, x_6, x_7, x_8).$$

This produces a scalar or vector measurement that is stable under small perturbations. If the Q_i are chosen to be projectively invariant or scale-normalized, then the measurement survives projection.

For example, a normalized quad could be

$$\widehat{Q}(a, b, c, d) = \frac{(a-c)^2 + (b-d)^2}{a^2 + b^2 + c^2 + d^2 + \eta},$$

where $\eta > 0$ prevents division by zero. Then $\widehat{Q}$ is insensitive to global scaling, which is useful in projective geometry.

A triple version is

$$\widehat{\mathcal{TQ}} = \widehat{Q}_1 + \widehat{Q}_2 + \widehat{Q}_3.$$

This gives a projectively stable local signature.

10. Recovering $O(n^3)$ Through Triple-Quad Sampling

Let the data field after folding be indexed as $X_{i,j,\ell}$, where

$$1 \le i, j \le n, \qquad 1 \le \ell \le 8.$$

The three folds give only 8 physical layers, so by themselves they give $O(8n^2) = O(n^2)$ storage.

To recover $O(n^3)$, we introduce a depth or filtration index $k = 1, \dots, n$. At each depth level k, we evaluate a triple-quad measurement:

$$\mathcal{TQ}_{i,j,k} = \mathcal{TQ}\left(X^{(k)}_{i,j,\ell}\right).$$

Then the full measurement field is

$$\{\mathcal{TQ}_{i,j,k} : 1 \le i, j, k \le n\}.$$

The number of measurements is

$$n \cdot n \cdot n = n^3.$$

Thus,

triple-quad measurements over an n-level projection filtration produce $O(n^3)$ structure.

This is the precise way the 3-folded packet leads to cubic complexity. The folds provide the local layer geometry; the filtration supplies the third asymptotic dimension.

11. Reverse Engineering Formula

Let

$$Z_{i,j,k} = \mathcal{TQ}_{i,j,k}$$

be the observed triple-quad measurement at planar site (i, j) and depth level k. We want to recover the hidden value field $V_{i,j,k}$.

A coherent inverse model is:

$$V^* = \arg\min_V \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} |\mathcal{TQ}(F_3(V))_{i,j,k} - Z_{i,j,k}|^2 + \lambda_1 R_{\text{smooth}}(V) + \lambda_2 G_\phi(V) + \lambda_3 R_{\text{top}}(V).$$

Here:

- $R_{\text{smooth}}(V)$ enforces geometric smoothness,
- $G_\phi(V)$ enforces golden anti-resonance,
- $R_{\text{top}}(V)$ enforces consistency with topological features.

A possible topological regularizer is

$$R_{\text{top}}(V) = d_{\text{B}}\left(\text{Dgm}(V), \text{Dgm}_{\text{target}}\right)^2,$$

where:

- $\text{Dgm}(V)$ is the persistence diagram of V,
- $\text{Dgm}_{\text{target}}$ is the desired topological signature,
- d_{B} is bottleneck distance.

So the reconstruction seeks a field whose projected triple-quad signatures and persistent topology match the observed data.

12. Conceptual Pipeline

The complete pipeline is:

1. **Start with a hidden cubic field**

$$V(i,j,k).$$

2. **Apply golden subdivision**
   Split planar indices according to

$$m \approx \frac{n}{\phi}, \qquad r \approx \frac{n}{\phi^2}.$$

3. **Fold three times**

   Produce an 8-layer packet:

$$L_3 = 8.$$

4. **Apply triple-quad measurements**
   Extract three coupled quadratic invariants from the folded packet.
5. **Project geometrically**
   Map the folded structure into projective coordinates:

$$[x : y : z : w].$$

6. **Build a topological filtration**
   Generate complexes

$$K_{\epsilon_1} \subseteq K_{\epsilon_2} \subseteq \cdots \subseteq K_{\epsilon_n}.$$

7. **Extract TDA signatures**
   Compute persistent features:

$$\beta_0, \beta_1, \beta_2.$$

8. **Solve inverse problem**
   Reconstruct the hidden value field:
   $$V^\ast=
   \arg\min_V
   \left|
   \mathcal{F}(V)-Y
   \right|^2
   +
   \text{regularization}.

Final Coherent Statement

Using a **triple-quad function** on a **3-folded packet** gives a coherent way to generate and analyze $O(n^3)$ structure from a folded $n \times n$ data manifold.

The key is that the three folds create an 8-layer local packet:

$$2^3 = 8,$$

while the projective-topological filtration contributes an additional depth index:

$$k = 1, \ldots, n.$$

Therefore the measurable interaction field has size

$$n \times n \times n,$$

so its complexity is

$$\boxed{O(n^3)}.$$

The triple-quad function acts as the local quadratic measurement mechanism. Projective geometry makes the measurements stable under scaling and folding. Topological data analysis extracts persistent global structure. Reverse engineering is then formulated as the inverse problem

$$\boxed{V^* = \arg\min_V \left[\|\mathcal{F}(V) - Y\|^2 + \lambda_1 R_{\text{smooth}}(V) + \lambda_2 G_\phi(V) + \lambda_3 R_{\text{top}}(V) \right].}$$

In short:

$$\boxed{3 \text{ folds} + \text{triple-quad invariants} + \text{projective TDA filtration} \Rightarrow O(n^3) \text{ reconstructable interaction geometry.}}$$